Omoroje Benjamin Alimele

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Education

Acadia University *May 2025 – Present*
B.Sc. Computer Science GPA**: 4.13/4.33**

- **Relevant Courses:** Intro to Python, Calculus I, Java, Computer Architecture.

Professional Experience

Extern **Remote**
AI-Powered Workflow Automation, Outamation *May 2025 – August 2025*

- Built AI-powered workflows to automate document classification and data extraction, using Natural Language Processing (NLP), Computer Vision, and Python-based pipelines (PyMuPDF, OCR techniques).
- Developed a retrieval system with LlamaIndex and Retrieval-Augmented Generation (RAG) to improve information search accuracy across mortgage documents, while evaluating open-source AI models for performance optimization.
- Synthesized technical findings by compiling a comprehensive analysis report on top AI models for document processing, highlighting key challenges, optimization strategies, and real-world deployment recommendations.

Projects & Outside Experience

Receipt Digitization & Management Platform *November 2025 – Present*

- Designed and developed a cross-platform receipt management tool (BillScan) using **Flutter**, **Firebase**, and **OCR libraries**, enabling automatic extraction and categorization of receipt data and reducing manual entry time by over **70%** for test users.
- Implemented a scalable backend workflow with **FastAPI**, **async OCR processing**, and secure cloud storage, improving document parsing accuracy by **25%** through optimized preprocessing and error-handling logic.
- Built an intuitive task and receipt dashboard with responsive UI components and local caching, increasing user engagement in early testing and supporting seamless cross-device syncing for rapid retrieval of financial records.

Vision-Based UI Automation Engine *November 2025 – Present*

- Developed a desktop application in **Python** to automate repetitive GUI-based tasks by recording and replaying user actions.
- Implemented computer vision using **OpenCV** to identify on-screen elements (icons, buttons, text fields), making the automation resilient to changes in layout and resolution.
- Engineered a backend with **FastAPI** and **SQLite** to manage and store user-created automation workflows via a REST API.

Raspberry Pi Display Signage System *October 2025*

- Collaborated with the Jodrey School's Computer Technician to create info displays that run on a **Raspberry Pi.**
- Diagnosed and fixed errors with getting the pictures from a remote server, downloading them locally, and putting them on display on a television.
- Optimized code to log the health status of the signage system to Github.

Volunteer & Leadership

First Year Representative
Acadia Computer Science Society *September 2025 - Present*

- Serve as the primary link between first-year Computer Science students and the ACSS executive team, communicating events, initiatives, and student feedback to enhance engagement and inclusion.
- Represent first-year interests at executive meetings and support society activities that foster community, collaboration, and participation among new students.

General Volunteer
Valley Harvest Marathon *October 2025*

- Assisted in cleaning up and clearing the racing area after the race had concluded.

Skills & Interests

Languages: Python, Java, Javascript, Typescript, SQL, HTML5, CSS, Dart
Developer Tools: Docker, Postman, Git
Libraries/Frameworks: FastAPI, Django, Flask, ReactJS, NextJS, PostgreSQL, Flutter